Assignment No. 09

Aim : Design a privacy-preserving Big Data system using data masking techniques.

step 1 : Verify Hadoop Environment

```
(base) admin1@nlpcomp10:~$ start-dfs.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [nlpcomp10]
nlpcomp10: Warning: Permanently added the ECDSA host key for IP address 'fd01::84d:e574:215d:3ff0' to the list of known hosts.
(base) admin1@nlpcomp10:~$
```

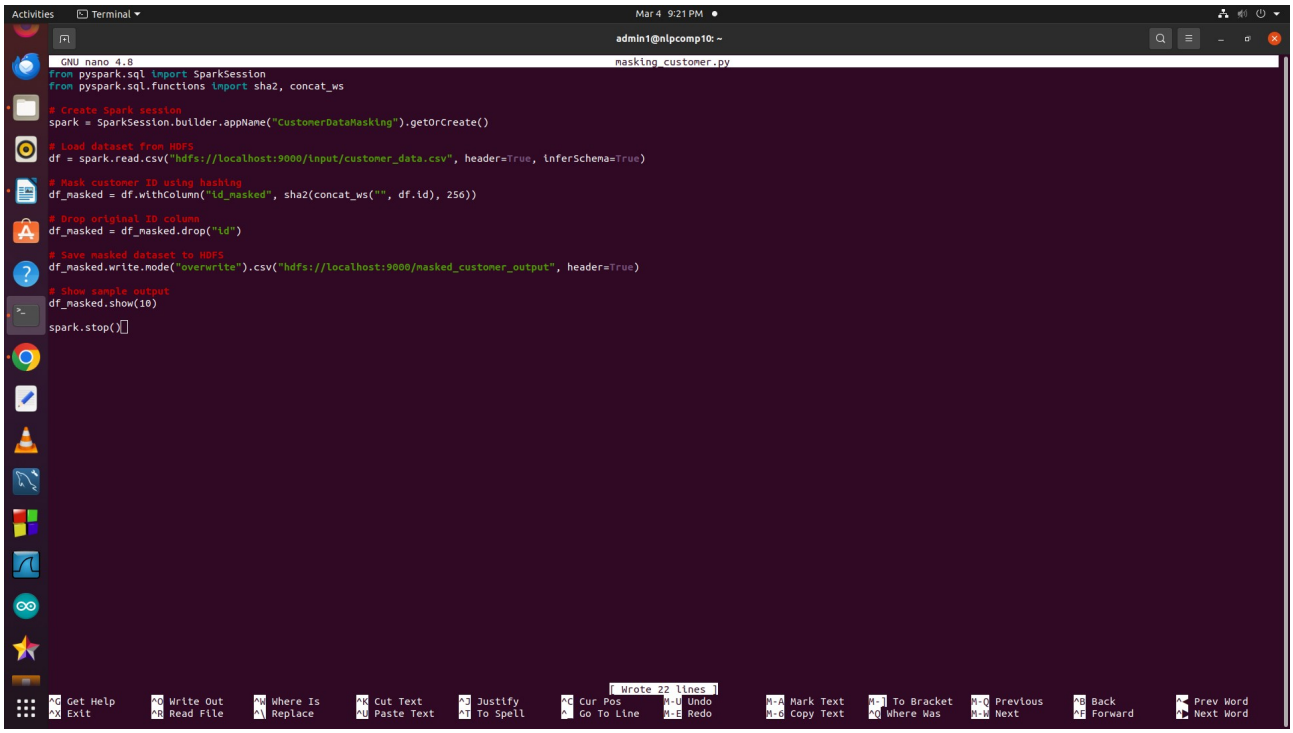
Step 2 — Prepare Input Dataset

```
(base) admin1@nlpcomp10:~$ ls /home/admin1/Downloads/Customer_Data.csv
/home/admin1/Downloads/Customer_Data.csv
(base) admin1@nlpcomp10:~$
```

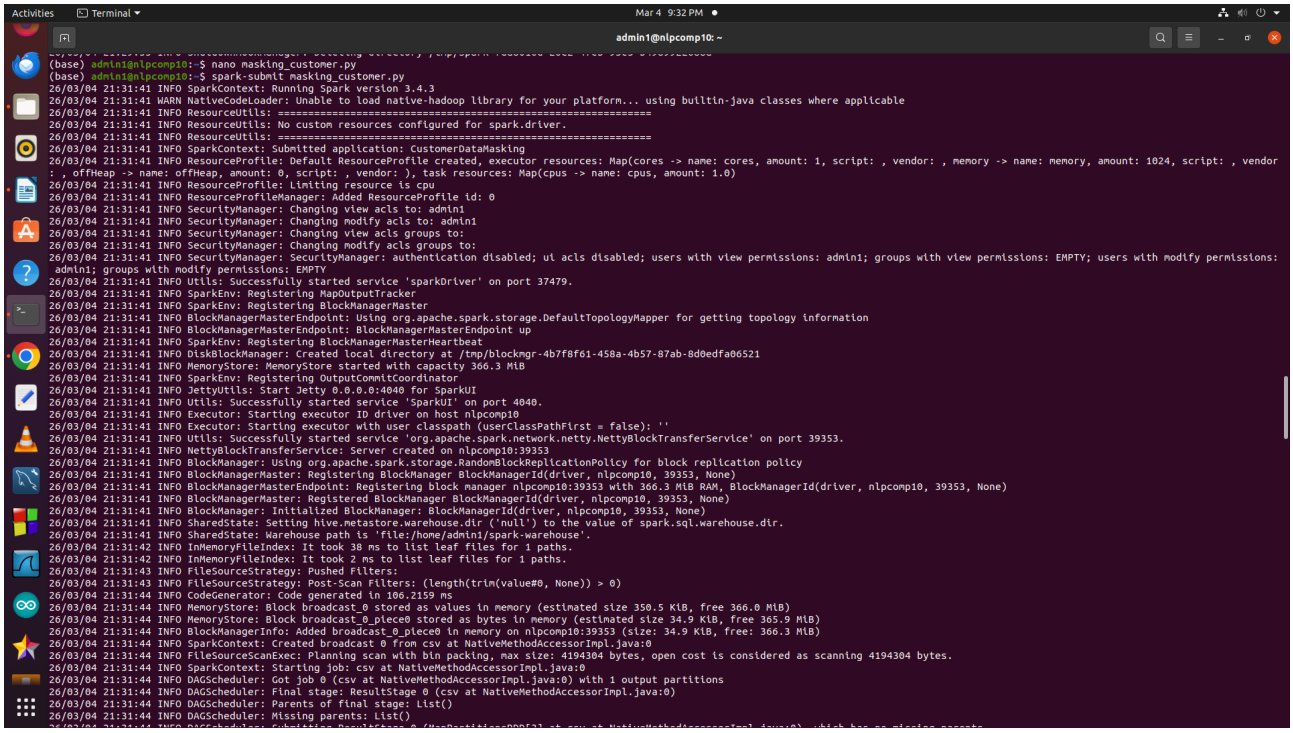
Step 3 — Create Input Directory in HDFS and upload dataset

```
/home/admin1/Downloads/Customer_Data.csv
(base) admin1@nlpcomp10:~$ hdfs dfs -mkdir -p /input
(base) admin1@nlpcomp10:~$ hdfs dfs -put /home/admin1/Downloads/customer_data.csv /input/
put: `/home/admin1/Downloads/customer_data.csv': No such file or directory
(base) admin1@nlpcomp10:~$ hdfs dfs -put /home/admin1/Downloads/Customer_Data.csv /input/
(base) admin1@nlpcomp10:~$
(base) admin1@nlpcomp10:~$ hdfs dfs -ls /input
Found 1 items
-rw-r--r--   1 admin1 supergroup     902876 2026-03-04 21:18 /input/Customer_Data.csv
(base) admin1@nlpcomp10:~$ hdfs dfs -cat /input/customer_data.csv | head -5
cat: `/input/customer_data.csv': No such file or directory
(base) admin1@nlpcomp10:~$ hdfs dfs -cat /input/Customer_Data.csv | head -5
CUSTID,BALANCE,BALANCE_FREQUENCY,PURCHASES,ONEOFF_PURCHASES,INSTALLMENTS_PURCHASES,CASH_ADVANCE,PURCHASES_FREQUENCY,ONEOFF_PURCHASES_FREQUENCY,PURCHASES_INSTALLMENTS_FREQUENCY,CASH_ADVANCE_FREQUENCY,CASH_
ADVANCE_TRX,PURCHASES_TRX,CREDIT_LIMIT,PAYMENTS,MINIMUM_PAYMENTS,PRC_FULL_PAYMENT,TENURE
C10001,40.900749,0.818182,95.4,0,95.4,0,0.166667,0,0.083333,0,0,2,1000,201.802084,139.509787,0,12
C10002,3202.467416,0.909091,0,0,0,6442.945483,0,0,0,0.25,4,0,7000,4103.032597,1072.340217,0.222222,12
C10003,2495.148862,1,773.17,773.17,0,0,1,1,0,0,0,12,7500,622.066742,627.284787,0,12
C10004,1666.670542,0.636364,1499,1499,0,205.788017,0.083333,0.083333,0,0.083333,1,1,7500,0,,0,12
cat: Unable to write to output stream.
(base) admin1@nlpcomp10:~$
```

Step 4 :Create Data Masking Script



Step 5 :Run Spark Data Masking Job



```
Activities   Terminal   Mar 4  9:32 PM
admin1@nlpcomp10: ~
26/03/04 21:31:45 INFO SparkContext: Created broadcast 5 from broadcast at DAGScheduler.scala:1540
26/03/04 21:31:45 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 2 (MapPartitionsRDD[13] at showString at NativeMethodAccessorImpl.java:0) (first 15 tasks are for partitions Vector(0))
26/03/04 21:31:45 INFO TaskSchedulerImpl: Adding task set 2.0 with 1 tasks resource profile 0
26/03/04 21:31:45 INFO TaskSetManager: Starting task 0.0 in stage 2.0 (TID 2) (nlpcomp10, executor driver, partition 0, NODE_LOCAL, 7925 bytes)
26/03/04 21:31:45 INFO Executor: Running task 0.0 in stage 2.0 (TID 2)
26/03/04 21:31:45 INFO FileScanRDD: Reading File path: hdfs://localhost:9000/input/Customer_Data.csv, range: 0-902876, partition values: [empty row]
26/03/04 21:31:45 INFO CodeGenerator: Code generated in 12.687105 ms
26/03/04 21:31:45 INFO Executor: Finished task 0.0 in stage 2.0 (TID 2). 3207 bytes result sent to driver
26/03/04 21:31:45 INFO TaskSetManager: Finished task 0.0 in stage 2.0 (TID 2) in 47 ms on nlpcomp10 (executor driver) (1/1)
26/03/04 21:31:45 INFO TaskSchedulerImpl: Removed TaskSet 2.0, whose tasks have all completed, from pool
26/03/04 21:31:45 INFO DAGScheduler: ResultStage 2 (showString at NativeMethodAccessorImpl.java:0) finished in 0.051 s
26/03/04 21:31:45 INFO DAGScheduler: Job 2 is finished. Cancelling potential speculative or zombie tasks for this job
26/03/04 21:31:45 INFO TaskSchedulerImpl: Killing all running tasks in stage 2: Stage finished
26/03/04 21:31:45 INFO DAGScheduler: Job 2 finished: showString at NativeMethodAccessorImpl.java:0, took 0.053385 s
26/03/04 21:31:45 INFO CodeGenerator: Code generated in 14.053144 ms
+------+----------+-----------------+---------+----------------+----------------------+-------------+-------------------+--------------------------+-------------------------------+----------------------+
-----------------+-----------+-------------+-----------+----------------+-----------------+------+
|CUSTID|    BALANCE|BALANCE_FREQUENCY|PURCHASES|ONEOFF_PURCHASES|INSTALLMENTS_PURCHASES|CASH_ADVANCE|PURCHASES_FREQUENCY|ONEOFF_PURCHASES_FREQUENCY|PURCHASES_INSTALLMENTS_FREQUENCY|CASH_ADVANCE_FREQUENCY|
CASH_ADVANCE_TRX|PURCHASES_TRX|CREDIT_LIMIT|   PAYMENTS|MINIMUM_PAYMENTS|PRC_FULL_PAYMENT|TENURE|
+------+----------+-----------------+---------+----------------+----------------------+-------------+-------------------+--------------------------+-------------------------------+----------------------+
-----------------+-----------+-------------+-----------+----------------+-----------------+------+
|C10001|  40.900749|         0.818182|     95.4|             0.0|                  95.4|          0.0|           0.166667|                       0.0|                       0.083333|                   0.0|
             0|          2|       1000.0| 201.802084|        139.509787|                   0.0|   12|
|C10002|3202.467416|         0.909091|      0.0|             0.0|                   0.0|  6442.945483|                0.0|                       0.0|                        0.0|                  0.25|
             4|          0|       7000.0|4103.032597|       1072.340217|              0.222222|   12|
|C10003|2495.148862|              1.0|   773.17|          773.17|                   0.0|          0.0|                1.0|                       1.0|                        0.0|                   0.0|
             0|         12|       7500.0| 622.066742|        627.284787|                   0.0|   12|
|C10004|1666.670542|         0.636364|  1499.0|          1499.0|                   0.0|   205.788017|           0.083333|                  0.083333|                        0.0|                0.083333|
             1|          1|       7500.0|        0.0|            null|                   0.0|   12|
|C10005| 817.714335|              1.0|     16.0|            16.0|                   0.0|          0.0|           0.083333|                  0.083333|                        0.0|                   0.0|
             0|          1|       1200.0| 678.334763|        244.791237|                   0.0|   12|
|C10006|1809.828751|              1.0|  1333.28|             0.0|                1333.28|          0.0|           0.666667|                       0.0|                       0.583333|                   0.0|
             0|          8|       1800.0| 1400.05777|       2407.246035|                   0.0|   12|
|C10007| 627.260806|              1.0|  7091.01|          6402.63|                 688.38|          0.0|                1.0|                       1.0|                        1.0|                   0.0|
             0|         64|      13500.0|6354.314328|        198.065894|                   1.0|   12|
|C10008|1823.652743|              1.0|    436.2|             0.0|                  436.2|          0.0|                1.0|                       0.0|                        1.0|                   0.0|
             0|         12|       2300.0| 679.065082|        532.03399|                   0.0|   12|
|C10009|1014.926473|              1.0|   861.49|          661.49|                  200.0|          0.0|           0.333333|                  0.083333|                       0.25|                   0.0|
             0|          5|       7000.0| 688.278568|        311.963409|                   0.0|   12|
|C10010| 152.225975|         0.545455|   1281.6|          1281.6|                   0.0|          0.0|           0.166667|                  0.166667|                        0.0|                   0.0|
             0|          3|      11000.0|1164.770591|        100.302262|                   0.0|   12|
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only showing top 10 rows

Masked Dataset
26/03/04 21:31:45 INFO FileSourceStrategy: Pushed Filters:
26/03/04 21:31:45 INFO FileSourceStrategy: Post-Scan Filters:
26/03/04 21:31:45 INFO CodeGenerator: Code generated in 19.072488 ms
26/03/04 21:31:45 INFO MemoryStore: Block broadcast_6 stored as values in memory (estimated size 350.4 KiB, free 364.8 MiB)
26/03/04 21:31:45 INFO MemoryStore: Block broadcast_6_piece0 stored as bytes in memory (estimated size 34.9 KiB, free 364.7 MiB)
26/03/04 21:31:45 INFO BlockManagerInfo: Added broadcast_6_piece0 in memory on nlpcomp10:39353 (size: 34.9 KiB, free: 366.1 MiB)
26/03/04 21:31:45 INFO SparkContext: Created broadcast 6 from showString at NativeMethodAccessorImpl.java:0
26/03/04 21:31:45 INFO FileSourceScanExec: Planning scan with bin packing, max size: 4194304 bytes, open cost is considered as scanning 4194304 bytes.
26/03/04 21:31:45 INFO SparkContext: Starting job: showString at NativeMethodAccessorImpl.java:0
26/03/04 21:31:45 INFO DAGScheduler: Got job 3 (showString at NativeMethodAccessorImpl.java:0) with 1 output partitions
```

```
Activities   Terminal   Mar 4  9:32 PM
admin1@nlpcomp10: ~
26/03/04 21:31:45 INFO FileScanRDD: Reading File path: hdfs://localhost:9000/input/Customer_Data.csv, range: 0-902876, partition values: [empty row]
26/03/04 21:31:45 INFO Executor: Finished task 0.0 in stage 3.0 (TID 3). 3925 bytes result sent to driver
26/03/04 21:31:45 INFO TaskSetManager: Finished task 0.0 in stage 3.0 (TID 3) in 14 ms on nlpcomp10 (executor driver) (1/1)
26/03/04 21:31:45 INFO TaskSchedulerImpl: Removed TaskSet 3.0, whose tasks have all completed, from pool
26/03/04 21:31:45 INFO DAGScheduler: ResultStage 3 (showString at NativeMethodAccessorImpl.java:0) finished in 0.019 s
26/03/04 21:31:45 INFO DAGScheduler: Job 3 is finished. Cancelling potential speculative or zombie tasks for this job
26/03/04 21:31:45 INFO TaskSchedulerImpl: Killing all running tasks in stage 3: Stage finished
26/03/04 21:31:45 INFO DAGScheduler: Job 3 finished: showString at NativeMethodAccessorImpl.java:0, took 0.021438 s
+----------+----------------+-------+----------+----------------+----------------------+-----------+-------------------+--------------------------+-------------------------------+----------------------+-------
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|    BALANCE|BALANCE_FREQUENCY|PURCHASES|ONEOFF_PURCHASES|INSTALLMENTS_PURCHASES|CASH_ADVANCE|PURCHASES_FREQUENCY|ONEOFF_PURCHASES_FREQUENCY|PURCHASES_INSTALLMENTS_FREQUENCY|CASH_ADVANCE_FREQUENCY|CASH_AD
VANCE_TRX|PURCHASES_TRX|CREDIT_LIMIT|   PAYMENTS|MINIMUM_PAYMENTS|PRC_FULL_PAYMENT|TENURE|       Masked_CUSTID|
+----------+----------------+-------+----------+----------------+----------------------+-----------+-------------------+--------------------------+-------------------------------+----------------------+-------
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|  40.900749|         0.818182|     95.4|       0.0|            95.4|                   0.0|          0.0|           0.166667|                       0.0|                       0.083333|                   0.0|
       0|          2|       1000.0| 201.802084|        139.509787|                   0.0|   12|3c17e0d32010a62cd...|
|3202.467416|         0.909091|      0.0|       0.0|             0.0|                   0.0|  6442.945483|                0.0|                       0.0|                        0.0|                  0.25|
       4|          0|       7000.0|4103.032597|       1072.340217|              0.222222|   12|b0740918f75e8a2d6...|
|2495.148862|              1.0|   773.17|    773.17|             773.17|                   0.0|          0.0|                1.0|                       1.0|                        0.0|                   0.0|
       0|         12|       7500.0| 622.066742|        627.284787|                   0.0|   12|5709f1f1160571cc4...|
|1666.670542|         0.636364|  1499.0|    1499.0|          1499.0|                   0.0|   205.788017|           0.083333|                  0.083333|                        0.0|                0.083333|
       1|          1|       7500.0|        0.0|            null|                   0.0|   12|8c068170506807891...|
| 817.714335|              1.0|     16.0|      16.0|            16.0|                   0.0|          0.0|           0.083333|                  0.083333|                        0.0|                   0.0|
       0|          1|       1200.0| 678.334763|        244.791237|                   0.0|   12|2c5ddc5a25ab3b343...|
|1809.828751|              1.0|  1333.28|       0.0|             0.0|                1333.28|          0.0|           0.666667|                       0.0|                       0.583333|                   0.0|
       0|          8|       1800.0| 1400.05777|       2407.246035|                   0.0|   12|d6141ccab08523939...|
| 627.260806|              1.0|  7091.01|    6402.63|          6402.63|                 688.38|          0.0|                1.0|                       1.0|                        1.0|                   0.0|
       0|         64|      13500.0|6354.314328|        198.065894|                   1.0|   12|56338a0c6a7c532f9...|
|1823.652743|              1.0|    436.2|       0.0|             0.0|                  436.2|          0.0|                1.0|                       0.0|                        1.0|                   0.0|
       0|         12|       2300.0| 679.065082|        532.03399|                   0.0|   12|45ff3fe4c5ffed0ef...|
|1014.926473|              1.0|   861.49|    661.49|          661.49|                  200.0|          0.0|           0.333333|                  0.083333|                       0.25|                   0.0|
       0|          5|       7000.0| 688.278568|        311.963409|                   0.0|   12|f5755eac41d76e488...|
| 152.225975|         0.545455|   1281.6|    1281.6|          1281.6|                   0.0|          0.0|           0.166667|                  0.166667|                        0.0|                   0.0|
       0|          3|      11000.0|1164.770591|        100.302262|                   0.0|   12|d0b81ff5891607b38...|
+----------+----------------+-------+----------+----------------+----------------------+-----------+-------------------+--------------------------+-------------------------------+----------------------+-------
---------+-----------+-------------+-----------+----------------+-----------------+------+-------------------+
only showing top 10 rows

26/03/04 21:31:45 INFO FileSourceStrategy: Pushed Filters:
26/03/04 21:31:45 INFO FileSourceStrategy: Post-Scan Filters:
26/03/04 21:31:45 INFO FileOutputCommitter: File Output Committer Algorithm version is 1
26/03/04 21:31:45 INFO FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
26/03/04 21:31:45 INFO SQLHadoopMapReduceCommitProtocol: Using output committer class org.apache.hadoop.mapreduce.lib.output.FileOutputCommitter
26/03/04 21:31:45 INFO CodeGenerator: Code generated in 12.028521 ms
26/03/04 21:31:45 INFO MemoryStore: Block broadcast_8 stored as values in memory (estimated size 350.4 KiB, free 364.3 MiB)
26/03/04 21:31:45 INFO MemoryStore: Block broadcast_8_piece0 stored as bytes in memory (estimated size 34.9 KiB, free 364.3 MiB)
26/03/04 21:31:45 INFO BlockManagerInfo: Added broadcast_8_piece0 in memory on nlpcomp10:39353 (size: 34.9 KiB, free: 366.1 MiB)
26/03/04 21:31:45 INFO SparkContext: Created broadcast 8 from csv at NativeMethodAccessorImpl.java:0
26/03/04 21:31:45 INFO FileSourceScanExec: Planning scan with bin packing, max size: 4194304 bytes, open cost is considered as scanning 4194304 bytes.
26/03/04 21:31:45 INFO SparkContext: Starting job: csv at NativeMethodAccessorImpl.java:0
26/03/04 21:31:45 INFO DAGScheduler: Got job 4 (csv at NativeMethodAccessorImpl.java:0) with 1 output partitions
26/03/04 21:31:45 INFO DAGScheduler: Final stage: ResultStage 4 (csv at NativeMethodAccessorImpl.java:0)
26/03/04 21:31:45 INFO DAGScheduler: Parents of final stage: List()
26/03/04 21:31:45 INFO DAGScheduler: Missing parents: List()
26/03/04 21:31:45 INFO DAGScheduler: Submitting ResultStage 4 (MapPartitionsRDD[21] at csv at NativeMethodAccessorImpl.java:0), which has no missing parents
26/03/04 21:31:45 INFO MemoryStore: Block broadcast_9 stored as values in memory (estimated size 224.1 KiB, free 364.1 MiB)
```
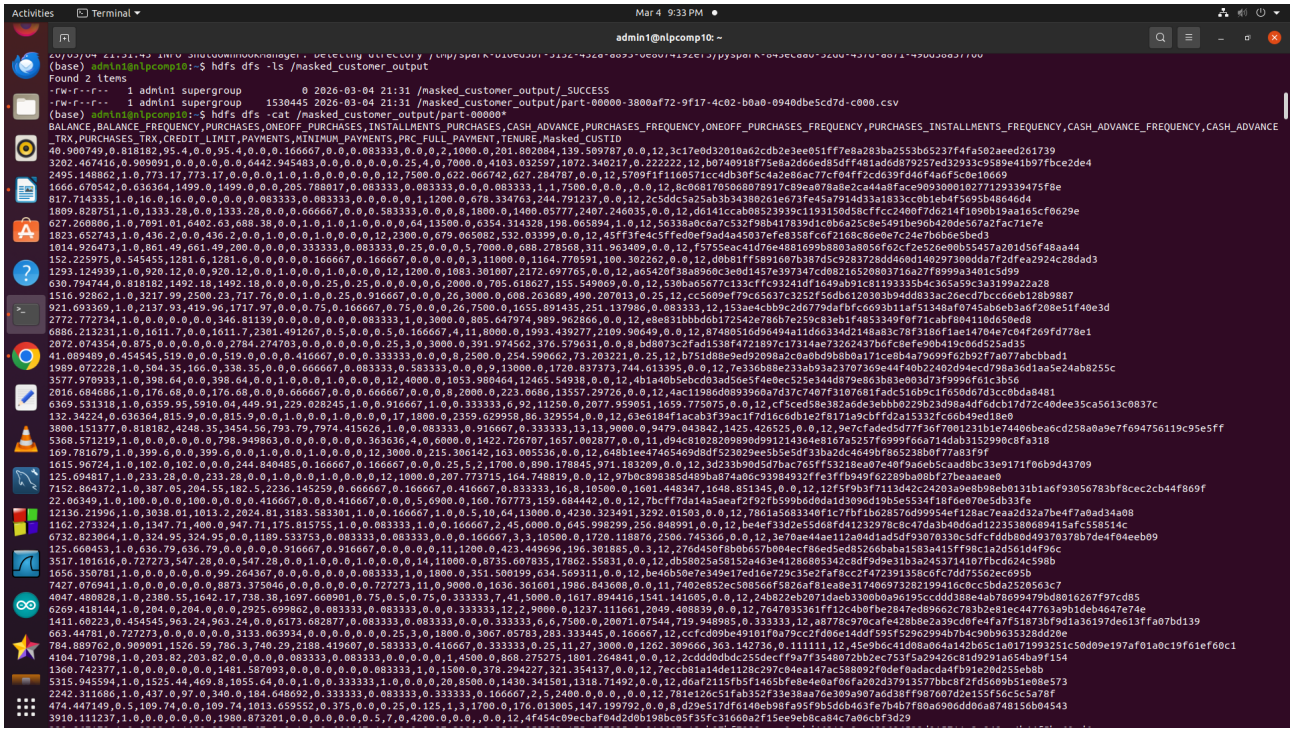
Step 6 :View Masked Dataset